



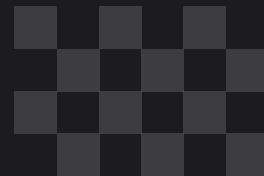
CYBER 207 · AI/ML FINAL PROJECT

Tuned for the Track

Detecting PII Before Prompts Leave the Gate

"Detect before you disclose."

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Why PII Detection Matters

Sensitive text can reach an AI tool before anyone checks it.



Names



Emails



Phone numbers



Addresses



ID / account numbers



...and more

As prompting and agentic workflows grow, sensitive text can leave the gate before anyone notices.



The Dataset Defined the Decision

- AI4Privacy provides multilingual text with token-level PII labels
- A single row can contain several different PII types
- We collapsed those annotations into one allow-or-review flag

8

PII spans in one synthetic
row

1

Document flag:
privacy-sensitive



Synthetic example: "My child faozzsd379223 (DOB: May/58)... ZIP 28170-6392... Email faozzsd379223@outlook.com."



The Question and Our Bet

Research question

How accurately and efficiently can specialized models screen text before it reaches an external LLM, compared with a general-purpose LLM?

Hypothesis

Models trained for this task will outperform prompted Phi-4 on PII detection and response time.



FERRARI

Task-specific

- Trained for this task
- Sparse or contextual inputs
- Validation-checked operating point



FORD

General-purpose

- Prompted, not fine-tuned
- Broad language capability
- No task-specific training



Four Models Enter the Race

Ford v. Ferrari, replayed as a model bake-off.

Each approach solves the same screening decision, but brings a different engine: sparse features, a neural network, contextual fine-tuning, or a prompted LLM.

Race question

Does the biggest model win?

Or the model tuned for this track?

Four cars on the track today:



Classical ML



Neural Network



DistilBERT



Phi-4 (LLM baseline)



One Dataset, One Frozen Split

- Source: AI4Privacy PII-Masking-400k
- Originally token-level PII labels, not binary and not an LLM-prompt dataset
- Collapsed to document-level: any PII token → Label 1, else Label 0
- Cleaning: removed 6 blank rows and 88 remaining exact duplicates; resplit with seed 42

Cleaned size

325,423

Class balance

67.4% PII / 32.6% safe

Train / Val / Test

260,338 / 32,542 / 32,543

Race Rules: trained models use the frozen split. Phi-4 uses a stratified 3,000-example test subset. We compare PII precision, recall, F1, macro F1, and confusion matrices.



How Classical ML and the MLP Saw Text



TF-IDF features

50,000 lowercase unigram + bigram features



Text features

10 text-shape features: length, digits, punctuation, HTML, and more



Pattern / regex features

8 flags: email, phone, URL, IP, dates, IDs, and more



Combined feature set

50,018 total features; fitted and scaled on training only



Feature ablation result

Combined features raised Logistic Regression PII F1 from 0.868 to 0.881 and reduced false negatives from 2,066 to 1,841.

DistilBERT and Phi-4 used text directly, not this matrix.



Who Won the Race?

PII F1 → higher is better

Logistic Regression

MLP ($t=0.02$)

DistilBERT

Phi-4*



PII recall is the security priority. PII F1 balances detection with the false alarms a real tool would create.



DistilBERT leads. *Phi-4 ran 3,000 rows; the other models ran the full 32,543-row test set.



Same Prompts, Very Different Results

Model	PII F1	FNs	Avg ms	Comparison
DistilBERT	0.979	41	2.4	Same 3,000
Phi-4	0.804	406	688.8	Same 3,000
Detection gap	+0.175	365 fewer	—	DistilBERT
Speed gap	—	—	287× faster	DistilBERT



On the exact same prompts, DistilBERT missed 41 PII examples; Phi-4 missed 406.



Phi-4 averaged 688.8 ms per prompt, about 287× slower than DistilBERT.



Where the Models Missed



DistilBERT false negatives

Dense numeric strings and placeholder-adjacent tokens; 447 misses on the full test set.



Phi-4 false negatives

Titled names, subtle disclosures, structured IDs/passwords, and non-English text; 406 of 3,000.



DistilBERT false positives

Ambiguous financial and vehicle IDs; 394 on the full test set.



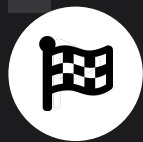
Phi-4 prompt tradeoff

False positives fell 547→382, while false negatives rose 326→406. The revision shifted the tradeoff.



Hard PII formats

Informal phones, nicknames, international addresses, IBAN-style codes, and dense-text zip codes.



Final Takeaway

“Detect before you disclose.”

- Specialized models outperformed prompted Phi-4 on this dataset-defined task
- DistilBERT led detection (PII F1 0.981) and averaged 2.4 ms per prompt
- Logistic Regression remained a credible lower-complexity option (PII F1 0.881; 1.55 ms)

For this task, model fit mattered more than model size.



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Thank you

Questions?

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